

UAS E2EE PIN Translation Developer Guide - 6.0.1-GA (AI)

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1. Preface - UAS E2EE PIN Translation Developer Guide

AccessMatrix Universal Authentication Server (UAS) provides a PIN translation feature as part of its End-to-End Encryption (E2EE) module. This document covers important details necessary for the E2EE PIN translation feature integration with business applications.

Intended Reader

This guide is intended for experienced application developers and security administrators.

The readers are assumed to have a good understanding about REST API and encryption technology in general.

Document Scope

The scope of this document covers the concept of ATM PIN translation and how to integrate it with business applications using REST APIs. It gives detailed integration use cases, along with REST API specifications and client sample codes in Java, Objective-C and Javascript.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

- [UAS Administration Guide](#)
- [UAS E2EEA Developer Guide](#)
- [AccessMatrix REST API Specification](#)

Specifically, for HSM integration information, see the following sections of the *UAS E2EEA Developer Guide*:

- [UAS E2EEA with SafeNet ProtectServer HSM Implementation](#)
 - [UAS E2EEA with Thales payShield 10K HSM Implementation](#)
 - [UAS E2EEA with SafeNet Payment HSM Implementation](#)
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2. Introduction to UAS E2EE PIN Translation

After reading this document, readers are expected to:

- Understand the concept of UAS E2EE ATM PIN translation.
- Understand the use case and workflow of UAS E2EE ATM PIN translation.
- Be able to implement UAS E2EE ATM PIN translation by integrating related REST APIs and client libraries.

Terminology

This table lists some common terms that are used in this document. It is not intended as a comprehensive definition for each item.

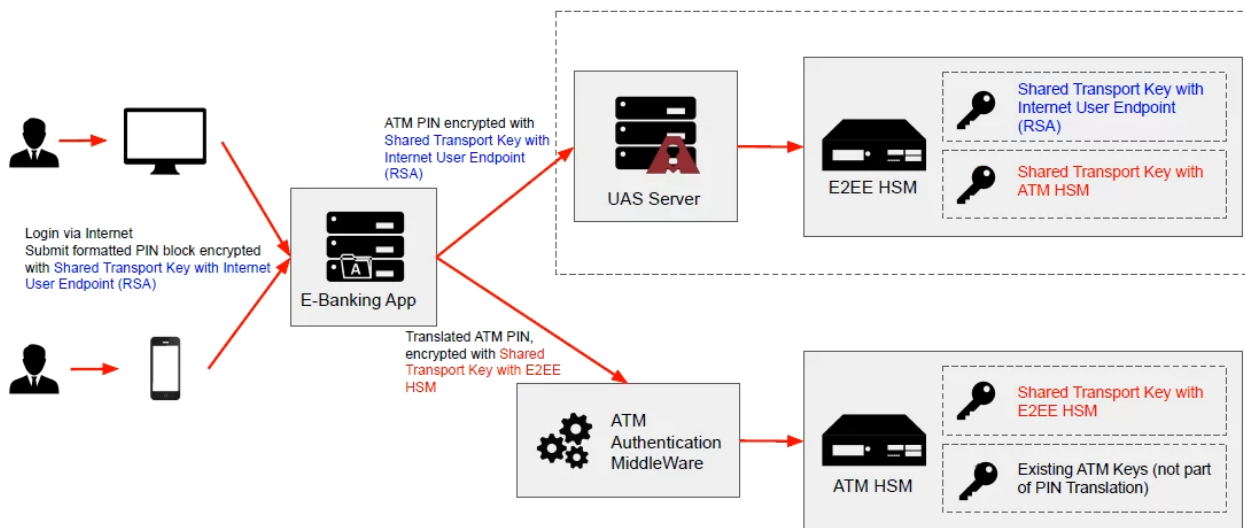
Term	Description
PIN Block	A PIN that is protected with public key encryption with special format. To protect against unauthorized disclosure or misuse.
PIN Format	The encoding format of the PIN block.
ISO-0	ISO 9564-1 Format 0. An ISO-0 PIN block format is equivalent to the ANSI X9.8, VISA-1, and ECI-1 PIN block formats and is similar to a VISA-4 PIN block format. The ISO-0 PIN block format supports a PIN from 4 to 12 digits in length. A PIN that is longer than 12 digits is truncated on the right. The first nibble (which identifies the block format) has the value 0.
TPK	Specific Thales payShield HSMs: A Terminal PIN Key (TPK) is a data-encrypting key which is used to encrypt PINs for transmission, within a local network, between a terminal and the terminal data acquirer.
ZPK	Specific Thales payShield HSMs: A Zone PIN Key (ZPK) is a data-encrypting key which is used to encrypt PINs for transmission, within a local network, between a terminal and the terminal data acquirer. Note: The choice to use TPK or ZPK depends on the customer.
LMK	Local Master Key, is used to encrypt operational keys used for encryption (like TPK and ZPK), etc. Local Master Keys are secret, internal to the HSM, and do not exist outside of the HSM.
PAN	This refers to the Primary Account Number.
PSE2	Thales ProtectServer HSM.
payShield	Thales payShield HSM.

Term	Description
EFT	Safenet/Luna EFT/EFT2 Payment HSM.

UAS E2EE ATM PIN Translation Concept

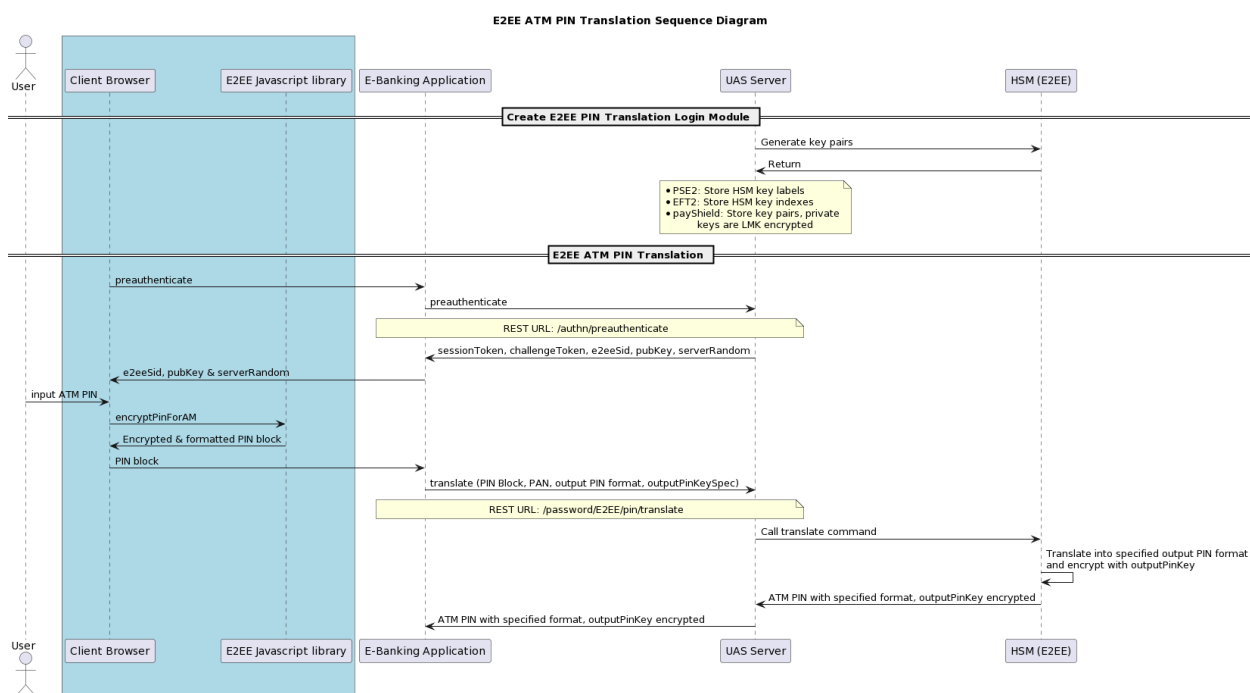
When it comes to payment process, a successful transaction usually involves various entities/nodes in the bank networks. In one case, the bank wants to use the same PIN for both ATM and Internet banking. During transmission of the PIN from user's endpoint (like web browser or mobile app) across those entities to verification node/HSM, the bank usually needs to translate the PIN to a particular PIN format. It is crucial to ensure that the whole ATM PIN translation process is securely encrypted end to end. UAS E2EE ATM PIN translation provides an extra layer of security by integrating E2EE technology.

Below is the component diagram of the UAS E2EE ATM PIN translation feature:



3. UAS E2EE PIN Translation Application Integration

Below is the sequence diagram of a typical use case of UAS E2EE ATM PIN translation:



- Use /authn/preauthenticate for preauthentication.
- Use /password/E2EE/pin/translate for PIN translation.

API Use Case

When you enter ATM PIN from the application front end (web or mobile), PIN translation is required. The ATM PIN needs to be translated securely inside the HSM (Hardware Security Module) to a specific format so that the ATM HSM can further process it. Below is the flow of ATM PIN translation:

1. A respective E2EE login module must be prepared (created) in UAS Server. Key pairs are generated by the HSM, and depends on the HSM:
 - **payShield** - The key pairs are stored in UAS, where the private keys are LMK encrypted.
 - **PSE2** - The key pairs are stored in HSM, and UAS stores HSM key labels.
 - **EFT** - The key pairs are stored in HSM.
2. User opens the application, that will call preauthentication to server (usually during app loading), to obtain a public key and random number.
3. User enters PIN such as "123456" into the application login screen.

-
4. The application will use the PIN to create a specially formatted PIN block by calling E2EE client library. The PIN block, depends on the HSM being used, is:
 - **payShield** - Public key encrypted with ISO-0 format.
 - **PSE2** - Public key encrypted with proprietary format.
 - **EFT** - Public key encrypted with proprietary format (EFT and PSE2 have the same format for this).
 5. The app front end submits the PIN block to the application back end.
 6. The application back end calls the UAS PIN translation REST API, passing in the encrypted PIN format, account number (if required for the output format), the desired output format and encryption parameters.
 7. UAS calls the HSM API/command to do the PIN translation, and returns the encrypted translated PIN to application back end.

REST Sample Codes

Refer to the [AccessMatrix REST API Specification](#) for API details and error codes.

E2EE Preauthentication Call	
URL: /authn/preauthenticate	
Request	
JSON	
<pre>{ "loginModuleId": "ps10k", }</pre>	
Response	
JSON	
<pre>{ "result": { "challengeToken": "00010FMPt9500Lsy_qmqtQrQ7fxFkO7617hBjfF_pbzFTp_4QyMh_- kJSGSpRY8ufQDeVdJSCxr0HRpkE82hPyHtw5g2phA", "authenticationCode": 0, "params": { "pubKeyIndex": "0003", } } }</pre>	

E2EE Preauthentication Call

```
{
  "expiredAt": "1600139613723",
  "oaepHashAlgo": "SHA256",
  "e2eeSid":
    "00010FMPt9500Lsy_qmqtQrQ7fxFkO76l7hBjfF_pbZFtp_4QyMh_-
    kJSGSpRY8ufQDeVdJSCxr0HRpkE82hPyHtw5g2phA",
  "loginModuleUUID": "oS",
  "serverRandom":
    "A7FDD51B519067749BBA395793546773",
  "pubKey":
    "F0D7EBAB8262B3A8C9D9A8F26C5444...10001"
},
  "amProcessingTimeMillis": 4
}
```



Notes:

- "loginModuleId" - E2EE login module Id
- "challengeToken" - challengeToken from server
- "oaepHashAlgo" - hash scheme of the login module
- "e2eeSid" - E2EE session id from the server
- "serverRandom" - server random number
- "pubKey" - public key from a server

E2EE Translate PIN API Call for Thales payShield HSM

URL: password/E2EE/pin/translate

Request

JSON

```
{
  "dynamicAgentSession": true,
  "loginModuleId": "ps10k",
  "E2EESID":
    "00010FMPt9500Lsy_qmqtQrQ7fxFkO76l7hBjfF_pbZFtp_4QyMh_-
    kJSGSpRY8ufQDeVdJSCxr0HRpkE82hPyHtw5g2phA",
  "RPIN": "00010FMPt9500Lsy_...A43CA36",
  "outputPinFormat": 0,
  "accountNumber": "123456789012",
  "outputPinKeySpec": {
    "keyLabel": "TPK",
    "keyValue":
      "T551528039FD706DC35C439AF90D2CF16D62717C0942C7300"
  }
}
```

Response

JSON

E2EE Translate PIN API Call for Thales payShield HSM

```
{
  "result": "B777C4F41555D539",
  "amProcessingTimeMillis": 176
}
```



Note: "keyValue" in "outputPinKeySpec":

- The sample request above contains a triple length key, which has the prefix "T".
- For double length keys, the prefix is "U".
- No prefix is required for single length keys.

Please refer to the Thales payShield documentation for more details.

E2EE Translate PIN API Call for Thales ProtectServer HSM

URL: password/E2EE/pin/translate

Request

JSON

```
{
  "dynamicAgentSession": true,
  "loginModuleId": "pse2login",
  "E2EESID": "0001SVEVQiBQNohHdoyOqh7oL25C1Py-
IpZyoB8N6Bw0esOMyWWQLnhCaFQ6Wj6GOcciUwdD-
I9FK2uZw0QMhOLO6w41rQU",
  "RPIN": "0001ONViK6053Lsy_...F13FD90",
  "outputPinFormat": 0,
  "accountNumber": "123456789012",
  "outputPinKeySpec": {
    "keyLabel": "BASED_ON_POLICY",
  }
}
```

Response

JSON

```
{
  "result": "A8934CF41555F007",
  "amProcessingTimeMillis": 175
}
```

E2EE Translate PIN API Call for Safenet/Luna EFT HSM

URL: password/E2EE/pin/translate

Request

E2EE Translate PIN API Call for Safenet/Luna EFT HSM	
JSON	<pre>{ "dynamicAgentSession": true, "loginModuleId": "EFT2login", "E2EESID": "0001SVEVQiBQNohHdoyOqh7oL25ClPy-IpZyoB8N6Bw0esOMyWWQLnhCaFQ6Wj6GOcciUwdD-I9FK2uZw0QMhOLO6w4lrQU", "RPIN": "0001ONViK6053Lsy_...F13FD90", "outputPinFormat": 1, "accountNumber": "123456789012", "outputPinKeySpec": { "keyLabel": "BASED_ON_POLICY", } }</pre>
Response	JSON <pre>{ "result": "A8934CF41555F007", "amProcessingTimeMillis": 175 }</pre>
 Note:	"RPIN" - The PIN block from application front end.

Client Sample Codes

Client libraries are provided in 3 languages: Java, Javascript and Objective-C, which covers both web and mobile use cases.

Javascript Sample

Use predefined functions in **ame2eea.js**:

E2EE ATM PIN Translation (Javascript) for Thales payShield HSM	
JavaScript	<pre>/* * Generate encrypted PIN with the desired format: * e2eeSid, pubKey, randomNumber and oaepHashAlgo are * returned from /authn/preauthenticate * If Login Module "Pin Block uses PAN/Token" is set to "PAN"</pre>

E2EE ATM PIN Translation (Javascript) for Thales payShield HSM

```
option, pass in accountNumber instead of randomNumber.
* format: use 'ISO-0' for ISO 9564-1 Format 0
*/
var pin = ame2eea.encryptPinForAM(e2eeSid, pubKey,
randomNumber /* accountNumber */, clearPIN, oaepHashAlgo,
'ISO-0');
```

E2EE ATM PIN Translation (Javascript) for Thales ProtectServer HSM and Safenet EFT HSM

JavaScript

```
/*
* Generate encrypted PIN with the desired format:
* e2eeSid, pubKey, randomNumber and oaepHashAlgo are
returned from /authn/preauthenticate
*
*/
var pin = ame2eea.encryptPinForAM(e2eeSid, pubKey,
randomNumber, clearPIN, oaepHashAlgo);
```

Objective-C/iOS Sample

Objective C APIs are provided in **libE2EEA.so**.

E2EE ATM PIN Translation (iOS) for Thales payShield HSM

C/C++

```
#import <E2EEA.h>
// create a new E2EEA object
E2EEA *e2eea = [[E2EEA alloc] init];
//If Login Module "Pin Block uses PAN/Token" is set to
"PAN" option, pass in accountNumber instead of randomNumber.
NSString *rpin = [e2eea encryptForTranslate_ISO_0: [e2eea
SHA256]
                session: e2eeSid
                password: @"1234"
                publicKeyValue: pubKey
                random: serverRandom
/*accountNumber*/];

if([e2eea getReturnCode]==0)
    NSLog(@"RPIN: %@", rpin);

// free the E2EEA object
[e2eea release];
```

E2EE ATM PIN Translation (iOS) for Thales ProtectServer HSM and Safenet EFT HSM	
C/C++	<pre> #import <E2EEA.h> // create a new E2EEA object E2EEA *e2eea = [[E2EEA alloc] init]; NSString *rpin = [e2eea encryptForLogin: [e2eea SHA256] session: e2eeSid password: @"1234" publicKeyValue: pubKey random: serverRandom]; if([e2eea getReturnCode]==0) NSLog(@"RPIN: %@", rpin); // free the E2EEA object [e2eea release]; </pre>

Java/Android Sample

Java APIs are provided in **e2eeAndroid.jar**, which are also applicable to Android development.

E2EE ATM PIN Translation (Java) for Thales payShield HSM	
Java	<pre> AxMxE2EEClient e2ee = new AxMxE2EEClient(); //If Login Module "Pin Block uses PAN/Token" is set to "PAN" option, pass in accountNumber instead of randomNumber. String encryptedPin = e2ee.encryptForTranslate_ISO_0(hashAlgorithmID, e2eeSid, pin, publicKey, serverRandom /*accountNumber*/) </pre>
E2EE ATM PIN Translation (Java) for Thales ProtectServer HSM and Safenet EFT HSM	
Java	<pre> AxMxE2EEClient e2ee = new AxMxE2EEClient(); String encryptedPin = e2ee.encryptForLogin(hashAlgorithmID, </pre>

E2EE ATM PIN Translation (Java) for Thales ProtectServer HSM and Safenet EFT HSM

<pre>e2eeSid, pin, publicKey, serverRandom)</pre>
